

Network Task - Sudir

Task Description:

1. Get me the IP address of a particular domain (guvi.in). How do I find my CPU/memory usage of my server?. Test the connectivity between 2 nodes?
2. I have deployed an application in guvi.com:9000, and logs show my app is running, but I'm unable to view the page. Check whether my port is open or not ?

Techstacks needs to be used :

- Shell (AWS, Gitbash, WSL, Vbox)

How do I submit my work?

- Push all your work files to GitHub (O/P screenshot images must).
- Submit your URLs in the portal.

Terms and Conditions?

- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.

NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

Solution:

Step 1:

Get me the IP address of a particular domain (guvi.in)

```
root@sonarqube:~# nslookup guvi.in
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   guvi.in
Address: 172.67.70.207
Name:   guvi.in
Address: 104.26.5.88
Name:   guvi.in
Address: 104.26.4.88
Name:   guvi.in
Address: 2606:4700:8d7a:d255:c13b:351:60b8:c78e
```

Step 2: How do I find my CPU/memory usage of my server?

```
root@sonarqube:~# top
top - 09:29:37 up 35 min,  2 users,  load average: 0.18, 0.36, 0.65
Tasks: 111 total,  1 running, 110 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us, 11.1 sy,  0.0 ni, 66.7 id,  0.0 wa,  0.0 hi, 22.2 si,  0.0 st
MiB Mem :  3403.7 total,  238.5 free,  2157.3 used,  1235.7 buff/cache
MiB Swap:  2048.0 total,  2047.8 free,  0.2 used,  1246.4 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24580   15124 10772 S   0.0   0.4   0:04.70  systemd
    2 root        20   0         0         0      0 S   0.0   0.0   0:00.00  kthreadd
    3 root        20   0         0         0      0 S   0.0   0.0   0:00.00  pool_workqueue_release
    4 root         0  -20         0         0      0 I   0.0   0.0   0:00.00  kworker/R-rcu_gp
    5 root         0  -20         0         0      0 I   0.0   0.0   0:00.00  kworker/R-sync_wq
```

htop

```

CPU[|||||] 21.8% Tasks: 36, 216 thr, 74 kthr; 1 running
Mem[|||||] 1.88G/3.32G Load average: 0.00 0.06 0.28
Swp[|] 248K/2.00G Uptime: 00:52:07

Main 77%
  PID USER      PRI  NI   VIRT   RES   SHR S  CPU% MEM%   TIME+  Command
2825 root        20   0   9144   5604   3940 R   1.6   0.2   0:00.08  htop
1268 sudir      20   0 18016   7696   5556 S   0.8   0.2   0:01.21  sshd-session: sudir@pts/0
2338 sudir      20   0 2680M   474M      0 S   0.8  13.9   0:00.50  /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.
2408 sudir      20   0 2566M   316M      0 S   0.8   9.3   0:12.73  /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.
    1 root        20   0 24580 15124 10772 S   0.0   0.4   0:04.76  /sbin/init
```

Free -h

```
root@sonarqube:~# free -h
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi         2.1Gi         236Mi        1.2Mi        1.2Gi         1.2Gi
Swap:          2.0Gi         252Ki         2.0Gi
```

1 second time interval provides 5 snapshots of the VM metric

```
root@sonarqube:~# vmstat 1 5
procs -----memory-----swap-----io-----system-----cpu-----
r b  swpd  free  buff  cache  si  so  bi  bo  in  cs  us  sy  id  wa  st  gu
2 0    252 241984 32112 1236128 0 0 672 701 1283 7 3 31 65 0 0 0
0 0    252 241984 32112 1236168 0 0 0 0 1180 446 0 22 78 0 0 0
0 0    252 241984 32112 1236168 0 0 0 0 1163 381 0 24 76 0 0 0
2 0    252 241984 32112 1236168 0 0 0 0 1154 434 0 22 78 0 0 0
0 0    252 241984 32112 1236168 0 0 0 0 1191 351 0 21 79 0 0 0
```

Step 3: Test the connectivity between 2 nodes?

```
root@sonarqube:~# ping 192.168.29.197
PING 192.168.29.197 (192.168.29.197) 56(84) bytes of data.
64 bytes from 192.168.29.197: icmp_seq=1 ttl=64 time=2.48 ms
64 bytes from 192.168.29.197: icmp_seq=2 ttl=64 time=1.95 ms
64 bytes from 192.168.29.197: icmp_seq=3 ttl=64 time=4.07 ms
64 bytes from 192.168.29.197: icmp_seq=4 ttl=64 time=2.18 ms
64 bytes from 192.168.29.197: icmp_seq=5 ttl=64 time=2.95 ms
^C
--- 192.168.29.197 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4008ms
rtt min/avg/max/mdev = 1.948/2.724/4.069/0.750 ms
```

```
root@sonarqube:~# telnet 192.168.29.197 22
Trying 192.168.29.197...
Connected to 192.168.29.197.
Escape character is '^]'.
SSH-2.0-OpenSSH_9.9p1 Ubuntu-3ubuntu3.1
```

Step 4: I have deployed an application in guvi.com:9000, and logs show my app is running, but I'm unable to view the page. Check whether my port is open or not?

In my case I have already deployed an application (sonarqube) in port 9000. To check as per the question asked in the task, I would troubleshoot using below commands.

First check service is up or not:

```
sudo@sonarqube:~$ systemctl status sonarqube.service
● sonarqube.service - Sonarqube Service
   Loaded: loaded (/etc/systemd/system/sonarqube.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-06-24 09:03:43 UTC; 46s ago
 Invocation: 4754ab9baf8d4dd3b944499ccb9fcell19
    Main PID: 2001 (java)
      Tasks: 59 (limit: 3981)
     Memory: 857M (peak: 857.2M)
        CPU: 41.831s
    CGroup: /system.slice/sonarqube.service
            └─2001 java -Xms8m -Xmx32m --add-exports=java.base/jdk.internal.ref=ALL-UNNAMED --add-opens=java
              └─2068 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=se
                └─2139 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Des.networkaddress.cache.ttl=60 -Des.network

Jun 24 09:04:20 sonarqube sonar.sh[2001]: 2025.06.24 09:04:20 INFO es[[o.e.p.PluginsService] loaded module
Jun 24 09:04:20 sonarqube sonar.sh[2001]: 2025.06.24 09:04:20 INFO es[[o.e.p.PluginsService] loaded module
Jun 24 09:04:20 sonarqube sonar.sh[2001]: 2025.06.24 09:04:20 INFO es[[o.e.p.PluginsService] loaded module
```

Check the status of app log, what caused the issue:

```
root@sonarqube:/home/sudir/sonarqube/sonarqube/sonarqube-25.5.0.107428/logs# tail sonar.log
2025.06.24 09:03:46 INFO app[[o.s.a.AppFileSystem] Cleaning or creating temp directory /home/sudir/sonarqube/sonarqube/sonarq
2025.06.24 09:03:46 INFO app[[o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:{}]
2025.06.24 09:03:47 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [/home/sudir/sonarqube/sonarqube/s
csearch]: /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli.script=./bin/elast
/server-cli -Des.path.home=/home/sudir/sonarqube/sonarqube/sonarqube-25.5.0.107428/elasticsearch -Des.path.conf=/home/sudir/sona
5.0.107428/temp/conf/es -Des.distribution.type=tar -cp /home/sudir/sonarqube/sonarqube/sonarqube-25.5.0.107428/elasticsearch/lib
r/sonarqube-25.5.0.107428/elasticsearch/lib/cli-launcher/* org.elasticsearch.launcher.CliToolLauncher
2025.06.24 09:03:47 INFO app[[o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
2025.06.24 09:04:59 INFO app[[o.s.a.SchedulerImpl] Process[es] is up
2025.06.24 09:04:59 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[WEB_SERVER] from [/home/sudir/sonarqube/sonarqube/sona
b/jvm/java-17-openjdk-amd64/bin/java -Djava.awt.headless=true -Dfile.encoding=UTF-8 -Djava.io.tmpdir=/home/sudir/sonarqube/sonar
temp -XX:OmitStackTraceInFastThrow --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.lang=ALL-UNNAMED --ad
-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED --add-exports=java.base/jdk.internal.ref=ALL-UNNAMED --add-opens=jav
--add-opens=java.base/sun.nio.ch=ALL-UNNAMED --add-opens=java.management/sun.management=ALL-UNNAMED --add-opens=jdk.management/co
-UNNAMED -Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError -Dhttp.nonProxyHosts=localhost|127.*|::1] -cp ./lib/sonar-applicatio
ir/sonarqube/sonarqube/sonarqube-25.5.0.107428/lib/jdbc/h2/h2-2.3.232.jar org.sonar.server.app.WebServer /home/sudir/sonarqube/s
428/temp/sq-process12252175603187189439properties
2025.06.24 09:05:48 INFO app[[o.s.a.SchedulerImpl] Process[web] is up
2025.06.24 09:05:48 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[COMPUTE_ENGINE] from [/home/sudir/sonarqube/sonarqube/
r/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.awt.headless=true -Dfile.encoding=UTF-8 -Djava.io.tmpdir=/home/sudir/sonarqube/s
428/temp -XX:OmitStackTraceInFastThrow --add-opens=java.base/java.util=ALL-UNNAMED --add-exports=java.base/jdk.internal.ref=ALL
e/java.lang=ALL-UNNAMED --add-opens=java.base/java.nio=ALL-UNNAMED --add-opens=java.base/sun.nio.ch=ALL-UNNAMED --add-opens=java
L-UNNAMED --add-opens=jdk.management/com.sun.management.internal=ALL-UNNAMED -Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError -
|127.*|::1] -cp ./lib/sonar-application-25.5.0.107428.jar:/home/sudir/sonarqube/sonarqube/sonarqube-25.5.0.107428/lib/jdbc/h2/h
n.CeServer /home/sudir/sonarqube/sonarqube/sonarqube-25.5.0.107428/temp/sq-process5329383188894505683properties
2025.06.24 09:06:11 INFO app[[o.s.a.SchedulerImpl] Process[ce] is up
2025.06.24 09:06:11 INFO app[[o.s.a.SchedulerImpl] SonarQube is operational
```

check the ports is open or not:

```
root@sonarqube:~# netstat -tulpn | grep 9000
tcp6      0 :::9000          :::*             LISTEN      2270/java
root@sonarqube:~# ss -tulpn | grep 9000
tcp       LISTEN 0      25             *:9000          *:~             users: (("java",pid=2270,fd=13))
```

If the port is not open add a rule to allow port in firewall (iptables):

`sudo iptables -A INPUT -p tcp --dport 9000 -j ACCEPT`

```
root@sonarqube:~# sudo iptables -A INPUT -p tcp --dport 9000 -j ACCEPT
root@sonarqube:~# sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination          tcp dpt:9000
ACCEPT     tcp  --  anywhere              anywhere             tcp dpt:9000
```

Then check the app is running or not:

```
root@sonarqube:/home/sudir/sonarqube/sonarqube/sonarqube-25.5.0.107428/logs# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7b:aa:6c brd ff:ff:ff:ff:ff:ff
    altname enx0800277baa6c
    inet 192.168.29.7/24 metric 100 brd 192.168.29.255 scope global dynamic enp0s3
```



